
Technical Documentation: FIWARE Data Space Connector (Research v5)

Project: Data Space Implementation using the marketplace

Environment: K3s Kubernetes on Ubuntu server (LTS 24.04) — Fontys ICT Netlab

Date: April 13, 2026

Table of contents

1. Architectural Scenarios..... 2

2. Core Working Mechanisms..... 2

 2.1 The Trust & Identity Handshake 2

 2.2 Orchestration & Marketplace Flow 2

3. Configuration & Optimization 3

 3.1 Network & Proxy Configuration..... 3

 3.2 Automated Deployment Patches..... 3

4. Troubleshooting & Known Behaviors 4

 4.1 Common Research Errors..... 4

 4.2 Verification Cheat Sheet 4

1. Architectural Scenarios

The deployment supports three operational modes, each progressively increasing in complexity:

- **Minimum Viable Dataspace (MVD):** A peer-to-peer setup where the Provider and Consumer interact directly using a shared Trust Anchor for identity verification.
 - **Central Marketplace:** Introduces a neutral orchestrating layer (did:web:fancy-marketplace.biz). The marketplace brokers transactions, handles product offerings, and triggers contract management notifications between participants.
 - **Gaia-X Profile:** An advanced compliance-focused profile (-Plocal,gaia-x) introducing a Gaia-X Registry, ovc:Constraint for ODRL policies, and specialized Verifiable Credentials (SD-JWT).
-

2. Core Working Mechanisms

2.1 The Trust & Identity Handshake

Authentication follows the Self-Sovereign Identity (SSI) model. Unlike traditional logins, this involves a multi-step exchange:

1. **DID Resolution:** Each participant hosts a did:web document. The Trust Anchor and Verifiers resolve these to find public keys.
2. **TIR Registration:** The Trusted Issuers Registry (TIR) must recognize specific DIDs (e.g., fancy-marketplace.biz) to allow them to "vouch" for users.
3. **Manual OID4VC Flow:** For research verification, credentials are acquired by:
 - Authenticating with Keycloak to get a bearer token.
 - Requesting a Credential Offer URI.
 - Exchanging a pre-authorized code for the final Verifiable Credential (VC) JWT.

2.2 Orchestration & Marketplace Flow

In the Marketplace scenario, the "Notification Chain" is the primary engine:

1. **Offering:** Provider registers a ProductOffering on the BAE (Business API Ecosystem).
 2. **Purchase:** Consumer places an order via the storefront.
 3. **Notification:** Upon order completion, the Marketplace Contract Management notifies the Provider's Contract Management.
 4. **Authorization:** The Provider's PAP (Policy Administration Point) is updated. **Crucial:** You must wait ~5s for the OPA (Open Policy Agent) to sync after this step before attempting data retrieval.
-

3. Configuration & Optimization

3.1 Network & Proxy Configuration

- **Squid Proxy:** Acts as the bridge (localhost:8888) to the Fontys Netlab environment.
- **Dynamic DNS:** Uses .nip.io to map the VM's internal IP (e.g., 192.168.120.15) to hostnames.
- **Kernel Dependencies:** Before deployment, the br_netfilter module must be loaded: `sudo modprobe br_netfilter`. Without this, CoreDNS will fail with UDP timeouts.

3.2 Automated Deployment Patches

- **Pre-Deployment Cleanup:** To prevent port conflicts, orphaned K3s containers must be cleared:

```
docker ps -a | grep k3s | awk '{print $1}' | xargs -r docker rm -f
```

- **Keycloak Dependency Bypass:** Replaces the wait-for-keycloak init-container with a no-op to prevent recursive deadlocks in versions 8.3.0–8.5.2.
 - **APISIX Data Plane Patch:** Modifies health probes to prevent crash-loops and changes service types to ClusterIP for internal stability.
-

4. Troubleshooting & Known Behaviors

4.1 Common Research Errors

- **The "Null Token" Bug:** If an OperatorCredential returns null after a purchase, the TIR likely hasn't updated.
 - *Verify TIR Health:* `curl -s http://tir.192.168.120.15.nip.io:8080/v4/issuers | jq .`
- **CoreDNS UDP Timeout:** If services cannot resolve each other.
 - *Fix:* `kubectl rollout restart deployment/coredns -n kube-system`
- **Duplicate Event delivery:** Harmless 500 errors in Contract Management logs are normal and do not impact the final policy state.

4.2 Verification Cheat Sheet

Goal	Command / Endpoint
Check Issued DIDs	<code>curl -s http://dsconfig.192.168.120.15.nip.io:8080/did</code>
Verify TIR Status	<code>http://tir.192.168.120.15.nip.io:8080/v4/issuers</code>
Marketplace Health	<code>https://fancy-marketplace.192.168.120.15.nip.io</code>
Keycloak Metadata	<code>[URL]/.well-known/openid-credential-issuer</code>
